Linnæus University
Sweden

"The Role of Digitalization in Decarbonization: A Multidimensional Analysis of Technological, Organizational, and Societal Impacts

Assignment three: Literature review



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Course Code:41K524
Word Count:

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Abstract

This article explores the intersection of digitalization and decarbonization, analysing how technological, organizational, and societal dimensions contribute to sustainability transitions. Drawing on empirical evidence from various sectors, it identifies digital tools as transformative enablers for reducing emissions and optimizing resource use. However, the study also highlights critical challenges, including energy demand, digital divides, and resistance to change within organizations and societies. This literature review provides actionable strategies for policymakers and practitioners to harness digitalization for sustainability while addressing inherent trade-offs.

Keywords: digitalization, decarbonization, sustainability, carbon reduction, digital business

Introduction

Description of the Problem Area

With growing environmental concerns and the urgent need to tackle climate change, global attention has shifted toward sustainable development. This transition demands innovative strategies to reduce carbon emissions, optimize resource use, and enhance resilience across various sectors. Digitalization has emerged as a transformative force capable of reshaping industries, economies, and societal behaviours. Its role is particularly crucial in driving decarbonization and improving efficiency in areas such as energy, logistics, urban development, and mobility.

Digital technologies like the Internet of Things (IoT), artificial intelligence (AI), blockchain, and data analytics offer exciting opportunities to reduce emissions, streamline operations, and advance sustainability efforts. These tools enable smarter monitoring, predictive insights, and decentralized systems, helping industries use resources more efficiently and manage energy better. However, their adoption isn't without challenges. Technologies such as AI and blockchain require significant amounts of energy, which can sometimes outweigh their environmental benefits. Additionally, IoT devices can generate electronic waste and increase dependence on rare earth materials (Mora et al., 2018). Integrating these technologies also brings complexities, such as finding the right balance between efficiency improvements and potential social concerns like job losses or privacy issues. To truly unlock the potential of digitalization for sustainability, it's essential to address these challenges and ensure these tools drive meaningful, long-term change. Digitalization’s intersection with Information Systems and Digital Business accelerates technological adoption and organizational transformation, enabling innovative business models and improving operational efficiencies. Despite these advancements, significant knowledge gaps persist in understanding how digital technologies can be effectively harnessed to achieve sustainability goals while mitigating their inherent trade-offs.

Previous literature reviews have explored the relationship between digitalization and decarbonization, often focusing on specific sectors like energy systems, shipping, or smart cities. These reviews provide valuable insights into how digital technologies, such as smart meters and AI, contribute to sustainability. They also highlight the role of policies and practices in fostering the adoption of digital tools and emerging trends like democratization and decentralization within energy systems. Despite these valuable contributions, many of these reviews are siloed, addressing only individual sectors or specific applications. A broader synthesis that connects findings across multiple domains is missing, limiting our understanding of digitalization’s overall impact on decarbonization.

Significance of the Study

This study seeks to address several knowledge gaps in the existing literature. First, many reviews focus heavily on technological impacts while overlooking the organizational and societal dimensions of digitalization. Issues such as public participation, digital divides, and the broader societal impacts of digitalization are often excluded. By addressing these aspects, this review provides a more holistic perspective on the intersection of digitalization and decarbonization. Additionally, existing reviews often rely on theoretical frameworks, with limited emphasis on

empirical evidence. This review prioritizes empirical studies to provide a grounded, data-driven understanding of real-world applications and outcomes.

Another limitation in current reviews is the lack of practical guidance for businesses and policymakers. While some reviews touch on policy implications, they rarely connect findings to actionable strategies. This review aims to bridge that gap by providing practical recommendations for decision-makers, ensuring that the insights are not only relevant but also actionable. Moreover, the field of digitalization and decarbonization is evolving rapidly. By incorporating the most recent studies, particularly from 2024, this review ensures that it reflects the latest developments, maintaining its relevance in a rapidly changing landscape.

The Aim of the Review

The aim of this literature review is to synthesize current research on the role of digitalization in sustainability, focusing on three key objectives:

1. Identifying the primary domains where digitalization facilitates decarbonization and resource efficiency.
2. Examining the trade-offs and challenges associated with digital technologies in sustainability applications.
3. Proposing actionable insights and future research directions to optimize the benefits of digitalization while minimizing its drawbacks.

Review Questions

To achieve the stated aim, the review addresses the following key questions:

1. How does digitalization contribute to decarbonization and resource efficiency across various sectors?
2. What are the primary challenges and trade-offs associated with deploying digital technologies for sustainability?
3. What frameworks or best practices exist for maximizing the environmental, social, and economic benefits of digitalization?
4. How can policymakers and organizations address the resource-intensive nature of digital solutions while achieving sustainability goals?

Contribution of the Review

This review offers a comprehensive synthesis of the relationship between digitalization and decarbonization, drawing insights from multiple sectors and dimensions, including technological, organizational, and societal aspects. By focusing on empirical evidence, it provides a clearer understanding of how digitalization drives decarbonization in practice. The review also contributes practical and policy-relevant findings, offering actionable recommendations for stakeholders seeking to implement digital strategies for sustainability. Furthermore, its incorporation of recent studies ensures the analysis is timely and relevant, addressing the rapidly evolving nature of this field.

By addressing these gaps and connecting findings across diverse domains, this literature review not only adds significant value to existing knowledge but also serves as a resource for researchers, practitioners, and policymakers. It aims to guide future efforts in leveraging digitalization as a tool for achieving decarbonization and broader sustainability goals.

Methods

This study aims to explore the impact of digitalization on decarbonization by examining existing empirical research on the subject. The primary objective is to identify how digital technologies, including digital business models and services, contribute to reducing carbon emissions and advancing sustainability goals such as carbon neutrality and net-zero emissions. By focusing on empirical studies, this research provides a data-driven understanding of the relationship between digitalization and decarbonization, rooted in real-world applications and measurable outcomes.

Search Procedure

The search procedure was carefully designed to identify studies that explore the intersection of digitalization and decarbonization. Keywords for digitalization, such as "digital business" and "digitalization," were chosen to capture research addressing how technology is reshaping industries and driving innovative business models. Keywords for decarbonization, including "decarbonization" and "carbon reduction," were selected to target studies focusing on environmental sustainability and carbon emissions reduction. This combination ensures the inclusion of studies examining the broader and more specific aspects of digitalization’s role in decarbonization.

The search strategy was constructed using Boolean operators for precision and inclusivity. The operator "AND" connected terms from different thematic areas, such as digitalization and decarbonization, ensuring the results addressed articles relevant to both domains. The operator "OR" included synonymous or closely related terms within each thematic area, ensuring comprehensive coverage. The final search strategy was as follows:

("digital business" OR "digitalization") AND ("decarbonization" OR "carbon reduction")

This query was applied to the titles and abstracts of articles to capture studies that delve deeply into the relationship between digitalization and decarbonization, avoiding articles where these terms are mentioned only in passing. The strategy aimed to uncover research exploring how digitalization contributes to decarbonization, focusing on measurable impacts and actionable insights.

Inclusion and Exclusion Criteria

To ensure the relevance and quality of the selected studies, specific inclusion and exclusion criteria were applied:

Inclusion Criteria:

1. The study must be a research article.
2. It must be empirical, focusing on real-world applications and measurable outcomes.
3. The article must examine the impact of digitalization on decarbonization.
4. Articles were included only if written in English and accessible as open access.

Exclusion Criteria:

1. Studies that were not empirical, such as policy papers or theoretical reviews.
2. Articles that were not classified as research articles, such as letters or editorial pieces.
3. Articles where the full text was not available in English.

Search Results and Selection Process

The initial search returned **56 articles** (Figure 1). Filtering for articles written in English reduced this number to **55** (Figure 2). Further refining the results to include only research articles narrowed the count to **26** (Figure 3). Finally, applying an open-access filter yielded a selection of **15 articles** (Figure 4). These articles represent the most relevant and recent research exploring the impact of digitalization on decarbonization.

The selection process was guided by the inclusion and exclusion criteria to ensure that only high-quality and relevant studies were included. A Prisma chart (Figure 5) summarizes the selection process, providing a visual representation of the filtering stages. See the Figure 1-4 in the Annex 1.

PRISMA Chart

The PRISMA chart below illustrates the systematic search and selection process, ensuring clarity and transparency at each stage:

1. **Records identified through database search:** The initial search returned 56 articles.
2. **Records after filtering for English-language articles:** After filtering, the number of articles was reduced to 55.
3. **Records screened:** Further refining the results to include only research articles narrowed the count to 26.
4. **Full-text articles assessed for availability:** Articles were filtered for open access reduced to 15.
5. **Full-text articles assessed for eligibility:** Articles were reviewed for relevance, focusing on their alignment with the inclusion and exclusion criteria.
6. **Final articles included:** A total of 12 articles met all criteria and were included in the review.

Summary of Selected Articles

The final selection comprises 12 empirical research articles are reflected in the Table 1 in Annex 2. The excluded articles due to eligibility are reflected in the Table 2 in Annex 2.

Data Summarization

The details of each article included in this study are thoughtfully compiled in Annex 3, offering a clear and organized overview. For each article, the annex answers key questions such as the title, the authors, and the journal or conference where it was published. It also dives into the research motivation, explaining why the study is essential and what knowledge gaps it addresses. You'll find the objectives, research questions, and concepts and theories used to guide the research. The annex outlines how the research was conducted, including the methodology, data collection, and analysis processes. It highlights the key findings, newly identified themes, and each article's contributions to the field. Limitations are also discussed, pointing out opportunities for future research. Finally, the annex includes personal reflections on what was learned, what was surprising, and what could have been improved. This structured summary makes it easy to grasp the essence and significance of each study within the broader context of this literature review.

Synthesis

This section brings together the themes identified in the conducted literature review, organizing them into a clear and practical theme matrix. It explores how digitalization and decarbonization intersect with sustainability and everyday organizational practices. By providing specific examples and insights, this synthesis shows how digital tools, policies, and frameworks shape areas like renewable energy, construction, and public administration. The themes cover a wide range of topics, from the practical challenges and opportunities of digitalization to its impact on society and the environment. They also highlight the crucial role that organizations and policies play in achieving sustainability goals. Each theme is broken into sub-themes, offering clear and actionable ideas for using digitalization to support decarbonization and innovation. Altogether, these themes paint a detailed picture of the relationship between digitalization and decarbonization. They address technological, organizational, and societal aspects, shining a light on both the possibilities and the challenges involved in creating a sustainable digitalization and low-carbon future. Common Themes Identified in the Literature are reflected in the Table 3 and Annex 4 (it includes the description and references).

Dimensions	**Themes**	**Sub-Themes**
Technological	Solar PV Performance	Dust Accumulation
		Maintenance
		Off-Grid Optimization
	Construction Digitalization	Prefabrication
		Real-Time Insights
	Data Centers & Energy Systems	Waste Heat Recovery
		Sector Coupling

	Infrastructure Maintenance	Precision Maintenance
		Monitoring
	Artificial Intelligence	Dual Role of AI
		Institutional Barriers
Organizational	Organizational Digitalization	Knowledge Management
		Technology-Sustainability Balance
	Energy Companies & Decentralization	Centralized Systems
		Regional Strategies
	Incumbent Strategies	Balancing Stability & Change
		Regional Dynamics
Societal	Public Administration	Digital Participation Platforms
		Renewable Energy in Rural Areas
	Societal Impacts of Digitalization	Energy Sufficiency & Skepticism
		Digital Divides
	Energy Democracy	Citizen Engagement
		Ownership Redefinition

Table 3. The conceptual framework driven from the study

Technological Dimension

1. **Solar PV Performance**

Dust Accumulation: Dust build-up on solar PV systems can severely impact their performance and energy output. Digital monitoring tools and predictive maintenance systems play a crucial role in reducing the need for manual interventions, thereby tackling this challenge efficiently. They identify inefficiencies early and ensure timely cleaning or adjustments, keeping systems running consistently and efficiently, making renewable energy more reliable. By lowering costs and supporting decarbonization efforts through enhanced energy efficiency, these technologies are a game-changer in the renewable energy sector (Ahmad et al., 2024).

'**Maintenance**: Maintaining off-grid solar PV systems has traditionally been resource-intensive, but IoT-enabled systems and advanced analytics are changing that. These tools allow for remote monitoring and automated problem detection, ensuring that maintenance happens proactively, exactly when and where needed. This proactive approach extends the life of PV systems, reduces downtime, and ensures they deliver maximum energy output. It is a clear example of how digital

tools contribute to sustainability by optimizing resource use and minimizing waste (Ahmad et al., 2024).

Off-Grid Optimization: Efficient energy management is critical for off-grid solar PV systems, especially in areas with limited resources. Data-driven solutions help analyse energy use patterns and predict demand, ensuring energy production and storage are used wisely. These innovations make off-grid systems more reliable and help communities reduce dependence on non-renewable energy sources. Most importantly, digitalization is driving urgent progress toward decarbonization goals by ensuring renewable energy is used to its fullest potential (Ahmad et al., 2024).

2. **Construction Digitalization**

Prefabrication: The combination of Building Information Modelling (BIM) and prefabricated components is revolutionizing the construction industry. By using detailed digital models to plan and design projects, prefabrication ensures that components are manufactured with high precision and fit seamlessly on-site. This approach dramatically reduces errors, material waste, and emissions, tackling sustainability challenges in an industry known for its heavy resource consumption. It also speeds up construction timelines, reducing the environmental impact of prolonged site activities like machinery use and transportation. Prefabrication shows how digital tools can make construction more innovative, efficient, and environmentally friendly (Piaia et al., 2024).

Real-Time Insights: Augmented Reality (AR) and BIM technologies are game changers in construction. They provide real-time insights that help workers stay on track with design plans, avoid mistakes, and use resources wisely. For instance, AR can project digital overlays on physical structures, guiding workers to align prefabricated components accurately. This reduces waste and energy use and helps projects run more smoothly. These tools bridge the gap between design and execution, showcasing best practices for sustainable construction while contributing to decarbonization goals (Piaia et al., 2024).

3. **Data Centers & Energy Systems**

Waste Heat Recovery: Data centres consume massive amounts of energy, making their sustainability a critical concern. However, digitalized energy systems address this issue by enabling waste heat recovery. The heat generated by servers can be captured and reused to warm nearby buildings or support industrial activities, significantly cutting down on overall energy demand. With advanced monitoring and control systems, these solutions ensure that heat recovery is efficient and reliable, directly contributing to decarbonization goals. This is an excellent example of how digital tools can turn a challenge into an opportunity, making energy use more innovative and sustainable (Li et al., 2024).

Sector Coupling: Imagine a world where energy from one sector seamlessly powers another. Digitalization makes this possible through sector coupling, where electricity, heating, and transportation systems work together. For instance, surplus renewable energy can charge electric vehicles or heat buildings, reducing waste and emissions. Digital tools like data analytics and automated controls make these connections efficient and impactful. This integrated approach

improves how resources are used and pushes us closer to a sustainable energy future (Li et al., 2024).

4. **Infrastructure Maintenance**

Precision Maintenance: Digital tools like Building Information Modelling (BIM) are changing the game in infrastructure maintenance. By offering detailed planning and real-time analysis, BIM ensures maintenance is done exactly when it's needed, saving resources and reducing emissions. For example, instead of overhauling an entire system, BIM helps pinpoint specific areas that need attention, minimizing waste and avoiding unnecessary disruptions. This precision extends the life of infrastructure and makes maintenance more efficient and sustainable (de Bortoli et al., 2024).

Monitoring: Real-time monitoring is taking infrastructure management to the next level. Using sensors and data analytics, these tools keep a close watch on the condition of roads, bridges, and buildings, identifying problems before they become serious. Think of it as a digital health check for infrastructure—catching minor issues early means fewer resources are wasted on significant repairs, and emissions from maintenance work are kept to a minimum. This shift from reactive to proactive management is a big step toward more sustainable infrastructure practices (de Bortoli et al., 2024).

5. **Artificial Intelligence**

Dual Role of AI: Artificial Intelligence is proving to be a game-changer in the push for sustainability. Not only does it stabilize existing systems, but it also drives innovative low-carbon solutions. For instance, AI-powered tools help optimize how energy is distributed, ensuring minimal waste and seamless integration of renewable sources. In industries, AI can spot inefficiencies in real-time, cutting down on unnecessary energy use and emissions. These applications highlight how AI is at the heart of digitalization's efforts to decarbonize and make systems more sustainable (John et al., 2024).

Institutional Barriers: However, tapping into AI's full potential isn't without challenges. Limited access to high-quality data makes it harder for AI to deliver precise results, while resistance to change within organizations slows down its adoption. Many companies also lack the infrastructure and expertise to implement AI-driven solutions effectively. Overcoming these hurdles requires better policies, improved data-sharing systems, and initiatives that encourage organizations to embrace AI. With the right support, AI could transform the way we approach sustainability (John et al., 2024).

Organizational Dimension

1. **Organizational Digitalization**

Knowledge Management: Effective knowledge management within organizations plays a pivotal role in reducing carbon emissions by fostering a culture of sustainability. Digital tools such as collaborative platforms and knowledge-sharing systems enable organizations to disseminate best practices,

streamline processes, and implement sustainable solutions more efficiently. For example, sharing data-driven insights on energy-saving practices across departments can significantly reduce resource waste and emissions. This shift in organizational culture, supported by digitalization, highlights actionable strategies for leveraging knowledge to achieve decarbonization goals (Jackson & Hodgkinson, 2024).

Technology-Sustainability Balance: The intersection of technology and sustainability requires organizations to strike a delicate balance. While digital tools drive efficiency and innovation, their implementation can sometimes lead to unintended consequences, such as increased energy consumption. Organizations must adopt holistic strategies that integrate technological advancements with environmental responsibility. For instance, choosing energy-efficient digital infrastructure and conducting lifecycle assessments of technologies can ensure that digitalization supports long-term decarbonization without exacerbating environmental challenges (Jackson & Hodgkinson, 2024).

2. **Energy Companies & Decentralization**

Centralized Systems: Digitalization is pushing energy systems toward decentralization, but big energy companies often stick to centralized models. These traditional setups shape how energy is produced and distributed, which can make it harder to adopt localized, sustainable solutions. For example, centralized grids might struggle to include small-scale renewable energy sources like local wind or solar projects. This shows how important it is to use digital tools that can connect centralized and decentralized systems, making energy production more flexible and sustainable (Kattirtzi et al., 2024).

Regional Strategies: Where a company operates and how it approaches strategy can make a big difference in achieving sustainability. Factors like local regulations, access to renewable resources, and regional energy demands all influence how effectively energy companies use digital tools to decarbonize. For instance, companies in regions with strong incentives for renewable energy are more likely to adopt digital technologies to improve efficiency and cut emissions. This balance between regional dynamics and company strategy highlights how complex—but essential—it is to bring about change in the energy sector (Kattirtzi et al., 2024).

3. **Incumbent Strategies**

Balancing Stability & Change: For established companies, the journey toward decarbonization isn't straightforward. They need to find a way to maintain the stability of their current systems while also embracing innovative technologies. This balance is crucial because their existing infrastructure often ensures reliability, but new digital tools like predictive analytics or energy optimization systems are essential for reducing emissions. It's a trade-off that requires thoughtful planning and a willingness to adapt long-standing practices to meet sustainability goals (Baumgartinger-Seiringer, 2024).

Regional Dynamics: Where a company operates can make a big difference in how it adopts digital technologies for decarbonization. Local factors like access to renewable energy resources, government regulations, and market demands

shape how easily firms can integrate digital tools. For example, companies in regions with strong renewable energy policies may find it easier to adopt new technologies, while those in areas with limited infrastructure face more hurdles. By understanding and adapting to these regional dynamics, firms can better navigate the complexities of digitalization and contribute meaningfully to sustainability (Baumgartinger-Seiringer, 2024).

Societal Dimension

1. **Public Administration**

Digital Participation Platforms: Digital participation platforms are making it easier for communities to shape their energy future actively. These tools allow people to come together, propose renewable energy projects, and collaborate on how to make them happen. By putting decision-making power into the hands of local communities, these platforms help decentralize energy systems and make renewable energy initiatives more accessible. They also build trust and transparency, showing how digitalization can drive community-level action to support sustainability (Jakimowicz & Rzeczkowski, 2024).

Renewable Energy in Rural Areas: Digital tools are a game-changer for rural communities, where access to energy infrastructure is often limited. They enable the creation of small-scale renewable energy projects, like solar microgrids or wind turbines, tailored to local needs. These solutions make energy more sustainable and reduce reliance on non-renewable sources. By using digital platforms to optimize energy use and manage resources efficiently, rural areas can overcome infrastructure challenges and contribute to the global push for decarbonization (Jakimowicz & Rzeczkowski, 2024).

2. **Societal Impacts of Digitalization**

Energy Sufficiency & Skepticism: Many people are sceptical about using technology as a solution to sustainability challenges. Concerns about whether these tools are reliable, fair, or might cause new problems often stand in the way of adoption. To overcome this scepticism, it's essential to build trust through transparency and public engagement. For example, showing people how digital tools work and involving them in the decision-making process can make these technologies feel less like abstract concepts and more like practical, reliable solutions. Building this trust is key to ensuring that digital tools are embraced as part of the journey toward energy sufficiency and sustainability (Gantioler et al., 2024).

Digital Divides: Not everyone has equal access to the digital tools needed to support sustainable practices, and this digital divide is a major roadblock. Communities with limited internet access, outdated infrastructure, or insufficient technical knowledge are often left out of the benefits that digitalization can bring. Closing this gap means investing in digital infrastructure, providing education, and offering support to under-resourced areas. When everyone has access to these tools, digitalization can truly drive broad, equitable progress toward decarbonization and sustainability (Gantioler et al., 2024).

3. **Energy Democracy**

Citizen Engagement: Digital tools are making it easier for people to get involved in shaping the future of energy. Whether it's through apps that let users monitor their energy consumption, platforms for investing in renewable projects, or forums for public input, these technologies are empowering communities to take charge of their energy choices. By giving citizens a voice in energy transitions, digitalization strengthens trust, ensures policies reflect community needs, and speeds up decarbonization efforts. It's a win-win for democracy and sustainability (Judson et al., 2024).

Ownership Redefinition: Digitalization is changing the way energy ownership works. Tools like blockchain and online energy trading platforms make it possible for individuals and communities to own and manage renewable energy sources, like local solar farms or shared wind turbines. This creates exciting opportunities for people to be more directly involved in the energy economy. But it also comes with challenges, like navigating regulations and ensuring everyone has access to these digital platforms. By addressing these hurdles, we can make sure this shift in ownership benefits both communities and the planet (Judson et al., 2024).

Conclusion

This review examined how digitalization supports decarbonization across technological, organizational, and societal dimensions. Analysing key areas of impact, trade-offs, and actionable strategies provided a comprehensive look at how digital tools drive sustainability transitions. The findings highlight the transformative potential of digitalization in areas such as renewable energy, construction, and public administration. Digital technologies can improve energy efficiency, foster community engagement, and encourage innovation. However, barriers like digital divides, organizational inertia, and limited data access pose significant challenges to fully realizing these benefits. Addressing these barriers requires well-thought-out policies, inclusive strategies, and a focus on both opportunities and risks to maximize the impact of digitalization in achieving decarbonization goals.

Future Research Agenda

To build on these insights and address gaps in the field, future research should focus on:

1. **Integrating Dimensions of Digitalization**: Explore how technological advancements, organizational strategies, and societal impacts can work together to enhance decarbonization efforts (Kattirtzi et al., 2024; Baumgartinger-Seiringer, 2024).

2. **Global and Regional Perspectives**: Conduct studies in diverse geographical and socio-economic settings to better understand the role of digitalization in sustainability transitions, especially in regions currently underrepresented in research (Jakimowicz & Rzeczkowski, 2024; Gantioler et al., 2024).

3. **Sector-Specific Applications**: Deepen research into specific industries such as construction, transportation, and public administration to identify tailored strategies for decarbonization (Piaia et al., 2024; Judson et al., 2024).

4. **Policy and Governance Insights**: Examine how digitalization intersects with policymaking to create frameworks that promote transparency, equity, and accountability in energy transitions (John et al., 2024; Kattirtzi et al., 2024).

5. **Emerging Technologies**: Investigate the potential of cutting-edge technologies like AI, blockchain, and IoT to address challenges in energy justice and sustainability (Li et al., 2024; Ahmad et al., 2024).

6. **Behavioral and Cultural Dynamics**: Study how societal perceptions, adoption barriers, and trust-building initiatives impact the implementation of digital tools for sustainability (Gantioler et al., 2024; Jackson & Hodgkinson, 2024).

7. **Economic Perspectives**: Conduct cost-benefit analyses of digitalization initiatives to help stakeholders make informed decisions about their long-term value and viability (Li et al., 2024; Jackson & Hodgkinson, 2024).

Focusing on these areas, future research can offer practical and actionable guidance for policymakers, organizations, and communities. Such efforts will position digitalization as a pivotal force in advancing decarbonization goals, paving the way for a low-carbon and sustainable future. By fostering collaboration across disciplines and engaging diverse stakeholders, digitalization can help create a more equitable and environmentally responsible world.

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Annex 1. Selection process

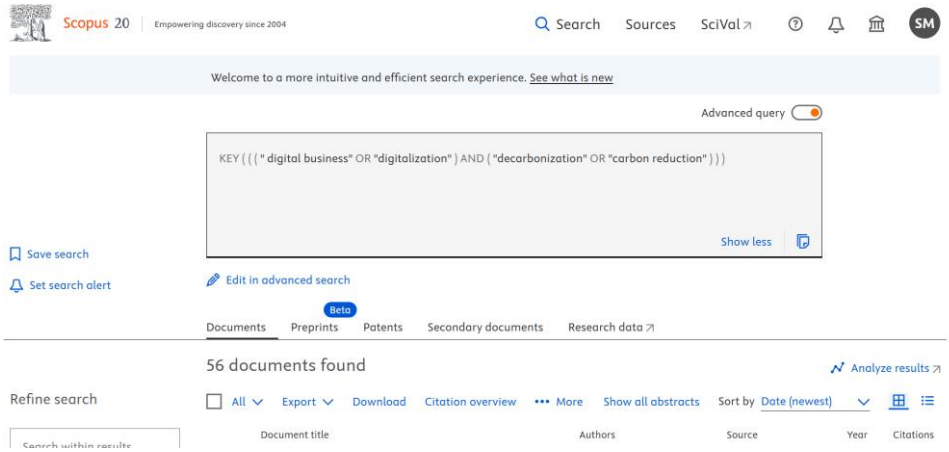


Figure 1. The initial search result with key words

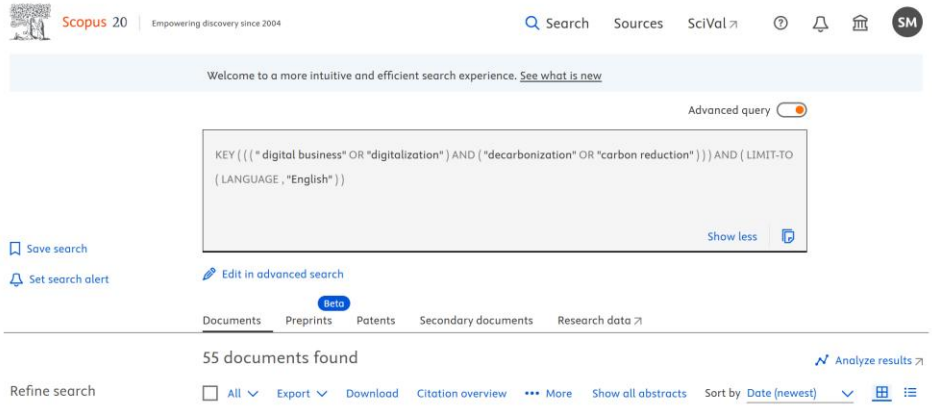


Figure2. The search results filtered with language choice of English only

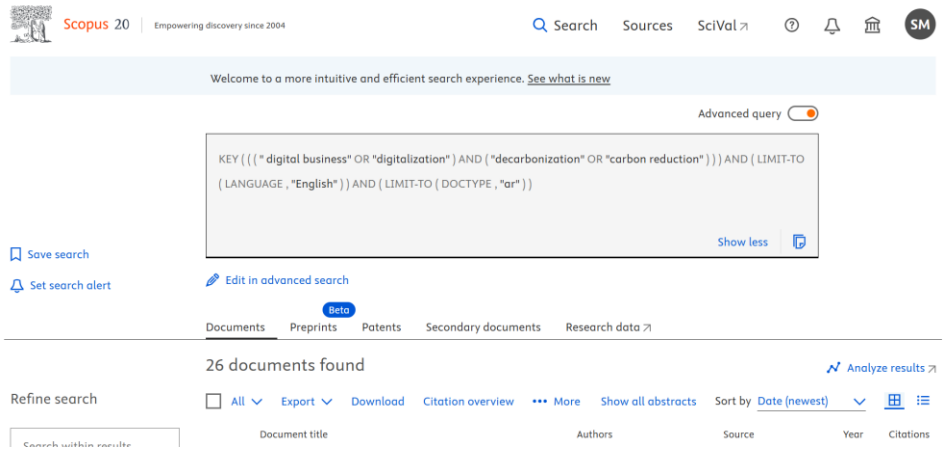


Figure 3. The search result filtered with language choice of English only and Article type of Research article only

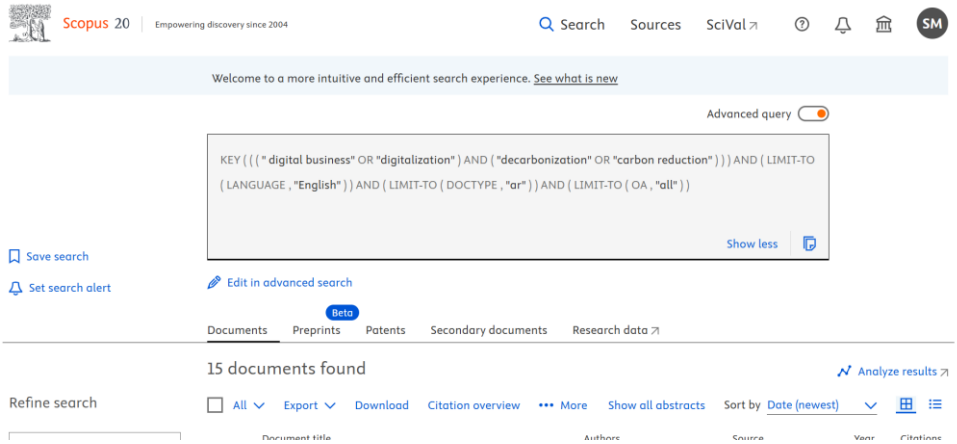


Figure 4. The search result filtered with open access only

Annex 2. The included and excluded articles in the study

Table 1. The included articles in the study						
No.	**Title**	**Authors**	**Year**	**Journal**	**Main Theme**	**Status**
1	Performance Analysis and Planning of Self-Sufficient Solar PV-Powered EV Charging Station	Ahmad et al.	2024	Renewable Energy	Solar PV performance and planning in digitalized systems.	Included
2	Advancing Decarbonization of the Construction Sector	Piaia et al.	2024	Journal of Construction and Building Materials	Digitalization's role in reducing emissions in construction.	Included
3	The Digitalization in Decarbonizing Electricity Systems	Heymann et al.	2024	Energy	Digital tools and electricity decarbonization.	Included
4	How Key-Enabling Technologies' Regimes Influence Decarbonization in the Steel Industry	John et al.	2024	Journal of Cleaner Production	Technological innovations in steel industry decarbonization.	Included
5	Energy Democracy: A Digital Future?	Judson et al.	2024	Energy Research & Social Science	Public participation in decarbonization through digitalization.	Included
6	The Role of Powerful Incumbent Firms in Shaping Regional Industrial Path Development	Baumgartinger-Seiringer	2024	Regional Studies, Regional Science	Organizational strategies in decarbonization.	Included

7	Digitalization and Deep Decarbonization in the Italian Smart Electricity Meter Roll-Out	Greiner et al.	2024	Journal of Energy Policy	Smart electricity meters and energy transition.	Included
8	Keeping a Lower Profile: How Firms Can Reduce Their Digital Carbon Footprints	Jackson & Hodgkinson	2024	Sustainability Science	Strategies for reducing digital carbon footprints.	Included
9	Electrification and Digitalization Effects on Sectoral Energy Demand	Li et al.	2024	Journal of Energy Studies	Energy demand impacts of digitalization and electrification.	Included
10	BIM Can Help Decarbonize the Construction Sector	de Bortoli et al.	2024	Journal of Cleaner Production	BIM technologies in construction sector decarbonization.	Included
11	The Impact of Public Administration Digitalization on Decarbonization	Jakimowicz & Rzeczkowski	2024	Energies	Digital public administration and its impact on emissions.	Included
12	Incumbents in Transition? The Role of the ‘Big Six’ Energy Companies in the UK	Kattirtzi et al.	2024	Energy Policy	Role of major energy companies in digitalization and decarbonization.	Included

Table 2. The excluded articles from the study

No.	**Title**	**Authors**	**Year**	**Journal**	**Main Theme**	**Reason of**	**Status**

						exclusion	
13	Presentation of the 5Ds in Energy Policy	Wagner & Götz	2024	Energies	Broader discussion of energy policy dimensions.	Conceptual	Excluded
14	Transformation of Industrial Enterprises: The Digital Aspect	Petrova et al.	2024	Industrial Policy Journal	General digital transformation of industries.	Full article unavaible in English	Excluded
15	The Digitalization, Dematerialization, and Decarbonization of the Global Economy	Smith & Martinez	2024	Historical Perspectives	Historical perspective on digitalization and decarbonization.	Conceptual	Excluded

Annex 3. Review included articles in the study

Feature of the article	Your comment
Title of the article	Advancing the Decarbonization of the Construction Sector: Lifecycle Quality and Performance Assurance of Nearly Zero-Energy Buildings
Author(s)	Emanuele Piaia 1 , Beatrice Turillazzi 2,* , Roberto Di Giulio 1 and Rizal Sebastian
Name of the journal/conference	sustainability
Research motivation stated by the author(s) (Why is the research important? What is the knowledge gap?)	The research highlights the construction sector's major role in energy consumption and emissions. With buildings using 40% of energy and producing 36% of greenhouse gases in the EU, there's an urgent need to address energy performance gaps during construction. The study fills the gap in real-time inspection methods for nearly zero-energy buildings (NZEBs), promoting sustainability and decarbonization.
Objective(s) & research question(s) (What is going to be done?)	The study aims to develop a real-time inspection methodology for NZEB construction using digital tools like BIM (Building Information Modeling) and AR (Augmented Reality). The goal is to reduce errors and ensure energy efficiency from design to lifecycle maintenance.
Concepts and theories used	NZEB principles and lifecycle quality assurance. Industry 4.0 technologies, including BIM, AR, and Digital Twins. Decarbonization and sustainable construction theories.
Methodology, data collection and data analysis (How was the research conducted?)	Steps: Site assessment, prefabricated component inspection, BIM modeling, and real-time AR-based self-instruction and inspection.

	Tools: Thermal and 3D laser scanning, BIM integration for design and quality checks. Analysis: Real-time monitoring to identify defects and optimize construction processes.
Key findings or results/ New identified themes (what has been found by the author(s)?)	Findings: Prefabricated solutions and digital tools reduced renovation timelines by 50% and improved energy savings by 70%. Thermal scanning identified issues like thermal bridges, ensuring better insulation and airtightness. Themes: Prefabrication and digitalization significantly enhance energy performance and reduce construction errors. AR and BIM empower workers with real-time insights, bridging the gap between design and execution. What has been found: The methodology ensures NZEB goals are met by reducing the gap between design intentions and actual performance. It highlights the importance of prefabricated components and advanced tools for real-time construction quality assurance.
Contributions to knowledge (What is the new knowledge/ How is the identified knowledge gap filled?)	The study introduces a practical framework for integrating advanced technologies in construction, addressing the performance gap in NZEBs. It provides a scalable solution for decarbonization goals, ensuring sustainability throughout the building lifecycle.
Limitations (possibility for future research, still existing knowledge gap)	Focuses mainly on EU scenarios, limiting global applicability.

	Economic analyses of long-term cost benefits are absent. Greater integration of user behavior in post-construction energy use could add depth.
Your own reflections (what have you learnt, what was already known to you, what was surprising, what could have been done in a better way?)	What I Learned: Digital tools can revolutionize construction, ensuring better energy efficiency and sustainability. What Was Surprising: The integration of AR and BIM not only reduced errors but also significantly shortened timelines. What Could Be Improved: Broader global studies and economic assessments could enhance the research's impact.

Feature of the article	**Your comment**
Title of the article	Performance analysis and planning of Self-Sufficient solar PV-Powered electric vehicle charging station in dusty conditions for sustainable transport
Author(s)	Ahmad, Fahad Faraza;Rejeb, Oussamaa;Kadir Hamid, Abdulb;Bettayeb, Maamarb;Ghenai, Chaoukia,
Name of the journal/conference	Transportation Research Interdisciplinary PerspectivesOpen AccessVolume 27September 2024 Article number 101214
Research motivation stated by the author(s) (Why is the research important? What is the knowledge gap?)	The research is motivated by the urgent need to address environmental challenges like rising energy demand and greenhouse gas emissions caused by petrol or diesel vehicles. It aims to explore clean energy solutions for EV infrastructure, focusing on the lack of studies analyzing the impact of dust accumulation on the performance of off-grid solar PV-powered EV charging stations.
Objective(s) & research question(s) (What is going to be done?)	The research aims to:Analyze and optimize the performance of a solar PV-powered EV charging station.Investigate the impact of dust accumulation over a year on PV efficiency.Achieve the "3Ds" of decarbonization, digitalization, and decentralization in energy and transportation.
Concepts and theories used	Sustainability concepts (decarbonization, digitalization, decentralization). Clean energy integration and performance analysis. Photovoltaic efficiency and dust accumulation theories.
Methodology, data collection and data analysis (How was the research conducted?)	Methodology: Performance analysis of a stand-alone solar PV-powered EV charging station. Data Collection: Real-time monitoring of energy production and consumption, weather data

	(irradiance, temperature, wind speed), and PV performance using smart sensors. Data Analysis: Monthly trends in PV production, energy consumption, and the effect of dust on performance ratio over a year.
Key findings or results/ New identified themes (what has been found by the author(s)?)	Key findings or results Dust accumulation caused a linear degradation in performance ratio from 77.92% to 27.30% in 218 days. The system operated at only 24.58% of its potential due to limited load demand. Increasing daily load demand or converting to a grid-connected system could maximize solar resource utilization. New identified themes: The significant impact of prolonged dust accumulation on solar PV performance. The role of cleaning and maintenance in maintaining PV efficiency. Opportunities for optimizing load management in off-grid PV systems. what has been found by the author(s)? Dust significantly affects PV efficiency, but after a certain accumulation threshold, its impact plateaus. The system effectively supports EV charging under dusty conditions but underutilizes available solar capacity.
Contributions to knowledge (What is the new knowledge/ How the identified knowledge gap is filled?)	The study provides new insights into how dust impacts the long-term performance of off-grid solar PV systems and identifies strategies to optimize their operation. By addressing the knowledge gap, it highlights the importance of regular maintenance and load demand

	optimization in achieving sustainable and efficient energy systems.
Limitations (possibility for future research, still existing knowledge gap)	The study has several limitations, including its narrow focus on the impact of dust while excluding factors like temperature and humidity, its single-location setup limiting generalizability, and its off-grid nature, which overlooks the potential benefits of grid integration. Additionally, low energy demand prevents the system from being used to its full potential. Future research could address these gaps by studying performance in diverse environmental conditions, exploring grid-connected systems, analyzing the economic aspects of maintenance in dusty areas, and developing innovative solutions like advanced cleaning technologies or dust-resistant materials.
Your own reflections (what have you learnt, what was already known to you, what was surprising, what could have been done in a better way?)	**What I Learned:** Dust accumulation significantly affects solar PV performance, particularly in desert regions, highlighting the need for optimization to maximize their environmental benefits. The "3Ds" framework—decarbonization, digitalization, and decentralization—emerges as a valuable tool for sustainable energy strategies. What Was Already Known: Solar energy is essential for sustainable transportation infrastructure, and dust is a well-known issue for PV systems. However, the detailed impact of dust over time was less familiar. What Was Surprising: The discovery that performance plateaus after substantial dust accumulation, with additional dust having minimal further impact, was unexpected. Additionally, the system's operation at only 24.58% of its capacity due to low load demand revealed a missed opportunity for optimization. What Could Have Been Done Better: Including an economic analysis to complement the technical findings, expanding the study to multiple geographic locations for broader applicability, and exploring alternative designs

	to mitigate dust effects without relying solely on cleaning could enhance the study's value.

Feature of the article	**Your comment**
Title of the article	The Impact of Public Administration Digitalization on the Decarbonization of the Economy
Author(s)	Aleksander Jakimowicz and Daniel Rzeczkowski
Name of the journal/conference	Energies
Research motivation stated by the author(s) (Why is the research important? What is the knowledge gap?)	The authors address the urgent need for decarbonization and highlight the untapped potential of digitalization in public administration to accelerate this process. They identify a gap in utilizing local government digital tools to foster renewable energy adoption and citizen participation in sustainable development.
Objective(s) & research question(s) (What is going to be done?)	The study aims to demonstrate how digitalizing municipal public administration can support decarbonization efforts, focusing on the Warmia and Mazury region in Poland. It investigates the role of digital platforms in enhancing community engagement and renewable energy adoption.
Concepts and theories used	Prosumption: A model where citizens actively participate in producing and consuming energy solutions. Wikinomics Framework: Highlights collaboration and innovation driven by digital tools. Decarbonization: Transition to low-carbon economies through sustainable practices.
Methodology, data collection and data analysis (How was the research conducted?)	Data Sources: Surveys from 1,821 participants in Warmia and Mazury. Analysis Methods: Correspondence analysis and clustering techniques to evaluate digital readiness and platform potential.

	Focus: Assessment of municipal websites as participation platforms for energy transition.
Key findings or results/ New identified themes (what has been found by the author(s)?)	Key Findings: Municipal digital tools are underdeveloped, limiting their ability to support renewable energy adoption. Public administration lags behind market sectors in implementing collaborative digital strategies. Community members are ready to engage but lack the digital platforms to do so effectively. New Identified Themes: The critical role of digital participation platforms in decarbonization. Barriers posed by insufficient digital infrastructure in public administration. The potential of small-scale renewable energy projects in rural regions. What Has Been Found: Digital public administration platforms, if well-developed, can significantly enhance decarbonization efforts by enabling community-driven renewable energy initiatives and improving administrative efficiency.
Contributions to knowledge (What is the new knowledge/ How the identified knowledge gap is filled?)	The study provides empirical evidence connecting public administration digitalization to decarbonization outcomes, emphasizing the need for integrating citizen-driven prosumption models in energy policy.
Limitations (possibility for future research, still existing knowledge gap)	Focused solely on Warmia and Mazury, limiting broader generalizability. Primarily theoretical; practical implementation remains untested.

	Future research could explore diverse regional contexts and implementation case studies.
Your own reflections (what have you learnt, what was already known to you, what was surprising, what could have been done in a better way?)	Learned: Effective digitization can empower local communities to contribute to decarbonization. Surprising: The disconnect between citizen readiness and administrative infrastructure. Improvements: Greater focus on real-world applications and broader geographic studies could enhance the study's impact.

Feature of the article	**Your comment**
Title of the article	Transformative disruptiveness or transition? Revealing digitalization and deep decarbonization pathways in the Italian smart electricity meter roll-out
Author(s)	Sonja Gantioler , Jessica Balest , Silvia Tomasi , Federico Voltolini , Nives DellaValle
Name of the journal/conference	Energy Research & Social Science
Research motivation stated by the author(s) (Why is the research important? What is the knowledge gap?)	The study addresses the urgent need to balance deep decarbonization and digitalization in energy systems. It highlights a gap in understanding how energy digitalization impacts justice and transformation, particularly through smart electricity meters (SEMs) in Italy.
Objective(s) & research question(s) (What is going to be done?)	The research aims to examine whether digital energy infrastructure, specifically SEMs, facilitates transformative disruption or merely incremental change toward decarbonization. It explores pathways shaped by societal values and justice concerns.
Concepts and theories used	Deep decarbonization and sustainability transition theories. Justice frameworks, including energy justice (distributional, procedural, and recognition). Bricolaging, emphasizing practical adaptation to systemic changes.
Methodology, data collection and data analysis (How was the research conducted?)	Approach: Qualitative research using abductive reasoning. Data Collection: Semi-structured interviews with 17 experts and 23 citizens, supplemented by content analysis.

	Analysis: Thematic coding using NVivo software, focusing on values, agency, and justice in energy digitalization.
Key findings or results/ New identified themes (what has been found by the author(s)?)	Key Findings: Two contrasting narratives emerged: energy as a commodity versus energy as a common good. SEMs can improve energy efficiency but may also exacerbate inequalities and limit individual agency. Identified Themes: Energy sufficiency and skepticism toward technological fixes. The role of societal values in shaping digitalization pathways. Vulnerabilities like energy poverty linked to digital divides. What Has Been Found: Digitalization holds potential for transformative decarbonization but risks reinforcing inequities without deliberate governance. SEMs contribute both opportunities and challenges to energy justice.
Contributions to knowledge (What is the new knowledge/ How the identified knowledge gap is filled?)	The study introduces a framework linking digitalization and decarbonization to justice and societal values. It emphasizes the need for inclusive governance to align technological innovations with equitable outcomes.
Limitations (possibility for future research, still existing knowledge gap)	Focuses on Italy, limiting global applicability. Insufficient exploration of economic impacts and policy-making dynamics.

	Limited data on citizen participation in decision-making.
Your own reflections (what have you learnt, what was already known to you, what was surprising, what could have been done in a better way?)	Learned: Digital tools can enhance energy systems but require careful integration to ensure fairness. Surprising: SEMs can unintentionally create barriers for vulnerable groups. Improvements: A broader geographic scope and more robust policy analysis could strengthen the findings.

Feature of the article	**Your comment**
Title of the article	Keeping a lower profile: how firms can reduce their digital carbon footprints
Author(s)	Thomas W. Jackson and Ian Richard Hodgkinson
Name of the journal/conference	JOURNAL OF BUSINESS STRATEGY
Research motivation stated by the author(s) (Why is the research important? What is the knowledge gap?)	The article highlights the overlooked environmental impact of digitalization, particularly the growing "digital carbon footprint" driven by data storage and dark data. The knowledge gap lies in how businesses can reduce emissions tied to digital practices while pursuing sustainability.
Objective(s) & research question(s) (What is going to be done?)	The study introduces the concept of digital decarbonization and explores how organizations can reduce their digital carbon footprint. It proposes strategies for better knowledge reuse to minimize energy waste from unnecessary data storage and duplication.
Concepts and theories used	Digital decarbonization: Reducing emissions associated with digital activities. Sociotechnical systems: Interactions between technology, people, and processes. Knowledge reuse: Efficient sharing and repurposing of organizational knowledge.
Methodology, data collection and data analysis (How was the research conducted?)	The authors used theoretical models and empirical data to analyze organizational practices in managing digital data and knowledge. They applied sociotechnical perspectives to propose pathways for reducing carbon footprints.
Key findings or results/	Key Findings:

New identified themes (what has been found by the author(s)?)	Digitalization contributes 4% of global greenhouse gas emissions. Over 55% of organizational data is "dark data"—unused but still stored, wasting energy. Knowledge reuse can significantly reduce digital carbon footprints by minimizing redundant data processes. New Identified Themes: The impact of "single-use knowledge" on carbon emissions. The importance of balancing technological advancements with sustainability. Encouraging cultural shifts within organizations to prioritize knowledge sharing. What Has Been Found: Digital practices, if unmanaged, can hinder sustainability efforts. However, adopting knowledge reuse strategies and effective digital governance can mitigate these effects, promoting a greener approach to data management.
Contributions to knowledge (What is the new knowledge/ How the identified knowledge gap is filled?)	The article introduces actionable insights into managing digital emissions, bridging the gap between sustainability and digitalization. It provides a framework for organizations to reduce carbon footprints through efficient knowledge management.
Limitations (possibility for future research, still existing knowledge gap)	Focused on theoretical and anecdotal data rather than broad empirical studies. Limited attention to sector-specific challenges and solutions. Future studies could explore the scalability of these strategies across industries.

Your own reflections (what have you learnt, what was already known to you, what was surprising, what could have been done in a better way?)	What I Learned: Digitalization is a double-edged sword—it enables progress but can harm the environment if unchecked. What Was Surprising: The sheer volume of unused "dark data" contributing to emissions. What Could Be Improved: The research could include real-world case studies to validate the proposed strategies and offer more specific examples of success.

Feature of the article	**Your comment**
Title of the article	Electrification and digitalization effects on sectoral energy demand and consumption: A prospective study towards 2050
Author(s)	Xiang Li, Dorsan Lepour , Fabian Heymann , François Maréchal
Name of the journal/conference	Energy
Research motivation stated by the author(s) (Why is the research important? What is the knowledge gap?)	The study tackles the challenge of balancing electrification and digitalization with sustainability goals, focusing on Switzerland. It highlights the gap in understanding how these trends impact sectoral energy demands and consumption at a national level, which is crucial for meeting carbon neutrality targets by 2050.
Objective(s) & research question(s) (What is going to be done?)	The research aims to model the long-term impacts of electrification and digitalization on energy demand and consumption across four sectors—residential, industry, mobility, and services. It seeks to answer how these trends influence energy system optimization and climate goals.
Concepts and theories used	Energy transition and decarbonization frameworks. Sector coupling and system optimization. The role of digital technologies (AI, IoT, and smart grids) in improving energy efficiency.
Methodology, data collection and data analysis (How was the research conducted?)	The study uses a hybrid top-down and bottom-up approach, analyzing historical data and modeling future energy trends. It incorporates macroeconomic drivers (population, GDP, temperature changes), system optimization (e.g., heat pumps), and the direct effects of digitalization.

Key findings or results/ New identified themes (what has been found by the author(s)?)	Key Findings: Electricity demand may increase by 20-32% by 2050, driven by electrification, while fuel consumption could drop by up to 95%. Digitalization could reduce total energy consumption by 10-30% through efficiency improvements. Data center heat recovery could satisfy 15% of national heating demand. New Identified Themes: The significant role of data center waste heat recovery in decarbonization. Digitalization as a dual driver of efficiency and increased electricity demand. Synergies between sector coupling and energy optimization. What Has Been Found: Electrification and digitalization are key to reducing energy consumption and emissions, but they require careful planning to mitigate risks such as increased electricity demand in ICT and mobility sectors.
Contributions to knowledge (What is the new knowledge/ How the identified knowledge gap is filled?)	The research offers a comprehensive model for understanding the interplay between electrification, digitalization, and energy system efficiency. It provides actionable insights for achieving carbon neutrality through strategic investments and technology integration.
Limitations (possibility for future research, still existing knowledge gap)	The study focuses on Switzerland, limiting its applicability to other contexts. Some variables, like building renovations and digital rebound effects, are underexplored.

	Broader stakeholder involvement could refine future scenarios.
Your own reflections (what have you learnt, what was already known to you, what was surprising, what could have been done in a better way?)	What I Learned: Digitalization has immense potential for energy savings, but its benefits depend on thoughtful integration into energy systems. What Was Surprising: Data centers could become energy suppliers through heat recovery, a game-changer for urban areas. What Could Be Improved: Incorporating case studies from diverse regions could enhance the study's global relevance and applicability.

Feature of the article	**Your comment**
Title of the article	BIM can help decarbonize the construction sector: Primary life cycle evidence from pavement management systems
Author(s)	Anne de Bortoli, Yacine Baouch, Mustapha Masdan
Name of the journal/conference	Journal of Cleaner Production
Research motivation stated by the author(s) (Why is the research important? What is the knowledge gap?)	The construction sector is a major contributor to carbon emissions, and stakeholders expect digitalization to drive decarbonization. However, there has been little quantifiable evidence of this impact, especially regarding the use of Building Information Modeling (BIM) for pavement management systems (PMS). This research fills that gap by assessing the carbon footprint of BIM in road construction and maintenance.
Objective(s) & research question(s) (What is going to be done?)	The study aims to quantify the carbon footprint of three road design-maintenance approaches, including one that integrates BIM for data-driven maintenance. It evaluates how BIM impacts greenhouse gas emissions and explores its potential for decarbonization.
Concepts and theories used	Life Cycle Assessment (LCA): A method to evaluate environmental impacts. BIM: Digital tools for managing construction and maintenance processes. Sustainability: Focus on reducing emissions and optimizing resource use.
Methodology, data collection and data analysis (How was the research conducted?)	The study uses LCA to compare three scenarios for road construction and maintenance over 30 years: Massive Design: Built to last with minimal maintenance.

	Progressive Design: Requires structural reinforcements based on initial planning. Data-Driven Maintenance: Uses BIM/PMS to adapt maintenance to actual conditions. Field data from France was analyzed, and carbon emissions were calculated for each scenario.
Key findings or results/ New identified themes (what has been found by the author(s)?)	Key Findings: Data-driven maintenance (Scenario 3) reduced emissions by up to 30% compared to traditional designs. BIM's direct emissions are negligible but provide indirect benefits through optimized maintenance. Using PMS resulted in better planning and fewer pavement failures. New Identified Themes: BIM’s potential to enable precision maintenance and minimize emissions. The need for real-time pavement condition monitoring. Limitations of traditional road maintenance methods in addressing sustainability goals. What Has Been Found: Integrating BIM for road maintenance improves sustainability by reducing unnecessary structural reinforcements and ensuring more efficient use of materials and energy.
Contributions to knowledge (What is the new knowledge/ How the identified knowledge gap is filled?)	This is the first study to provide quantifiable evidence of BIM’s environmental benefits in pavement management. It establishes a foundation for integrating digital tools into infrastructure decarbonization strategies.

Limitations (possibility for future research, still existing knowledge gap)	The study focuses on French infrastructure, limiting global applicability. Assumes static traffic growth and does not explore variable real-world conditions. Future studies could evaluate multi-functional BIM applications.
Your own reflections (what have you learnt, what was already known to you, what was surprising, what could have been done in a better way?)	Learned: Digital tools like BIM can revolutionize maintenance while cutting emissions. Surprising: BIM's direct emissions are minimal but have significant indirect benefits. Improvements: Including a broader geographic scope and real-time data could make the findings more comprehensive.

Feature of the article	**Your comment**
Title of the article	Incumbents in Transition? The Role of the ‘Big Six’ Energy Companies in the UK
Author(s)	Michael Kattirtzi, Ioanna Ketsopoulou, Jim Watson
Name of the journal/conference	Energy Policy
Research motivation stated by the author(s) (Why is the research important? What is the knowledge gap?)	The study investigates how the "Big Six" UK energy companies adapted to major disruptions like decarbonization, decentralization, and digitalization between 2008 and 2016. It aims to fill a gap in understanding the strategies these incumbents employed to navigate significant energy transitions.
Objective(s) & research question(s) (What is going to be done?)	The paper examines how the Big Six responded to these disruptive trends, focusing on their strategies in generation, retail, and networks. It questions whether they embraced innovation or clung to traditional approaches.
Concepts and theories used	Disruptive Innovation: Framework exploring how new technologies challenge established players. Creative Accumulation: How incumbents leverage existing assets to compete with new entrants. Decarbonization, Decentralization, and Digitalization: Key disruptive forces shaping energy systems.
Methodology, data collection and data analysis (How was the research conducted?)	Analyzed publicly available documents (e.g., reports, market data) and conducted 11 interviews with senior energy stakeholders. Thematic coding was applied using NVivo software to identify strategic responses to disruptions.

Key findings or results/ New identified themes (what has been found by the author(s)?)	Key Findings: The Big Six achieved progress in decarbonization, particularly through investments in renewables. Responses to decentralization were mixed, with a continued focus on centralization for most companies. Digitalization initiatives were limited, mainly concentrated on customer services with varying success. New Identified Themes: Centralized systems remain dominant despite decentralization trends. Parent companies' strategies significantly influenced UK subsidiaries' actions. Digitalization's potential for disruption is still underexplored. What Has Been Found: The Big Six demonstrated adaptability, especially in decarbonization, but their strategies primarily preserved existing business models rather than embracing transformative change.
Contributions to knowledge (What is the new knowledge/ How the identified knowledge gap is filled?)	The study offers insights into the role of incumbents in energy transitions, highlighting the interplay between policy, market incentives, and corporate strategies in shaping adaptation.
Limitations (possibility for future research, still existing knowledge gap)	The study focuses on the UK, limiting generalizability to other regions. It does not fully address long-term impacts of these strategies beyond 2016.

	Further research could explore comparisons with non-incumbent energy companies.
Your own reflections (what have you learnt, what was already known to you, what was surprising, what could have been done in a better way?)	Learned: Policy incentives strongly influence incumbent behavior, but maintaining the status quo often takes precedence over disruptive innovation. Surprising: Decentralization had a limited impact on most of the Big Six, contrary to global trends. Improvements: Broader case studies and recent data could provide a more comprehensive understanding of energy transitions.

Feature of the article	**Your comment**
Title of the article	Digitalization in Decarbonizing Electricity Systems – Phenomena, Regional Aspects, Stakeholders, Use Cases, Challenges, and Policy Options
Author(s)	Fabian Heymann, Tatjana Milojevic, Andrei Covatariu, Piyush Verma
Name of the journal/conference	Energy
Research motivation stated by the author(s) (Why is the research important? What is the knowledge gap?)	The study explores how digitalization can transform electricity systems while advancing decarbonization goals. It addresses a knowledge gap in understanding regional variations, stakeholders, and challenges in applying digital technologies for energy sustainability.
Objective(s) & research question(s) (What is going to be done?)	The research seeks to analyze digitalization in the electricity sector through a megatrend framework, identifying its benefits, challenges, and potential policy options to ensure its alignment with decarbonization objectives.
Concepts and theories used	Megatrend analysis framework Decarbonization and digitalization interconnectivity Stakeholder engagement and policy adaptation
Methodology, data collection and data analysis (How was the research conducted?)	Approach: A megatrend analysis framework comprising five stages, including phenomena identification, stakeholder analysis, and policy implications. Data Sources: Regional studies, stakeholder surveys, and global use case evaluations. Analysis: Cross-referencing global digitalization initiatives with their environmental and economic impacts.

Key findings or results/ New identified themes (what has been found by the author(s)?)	Key Findings: Digitalization improves system efficiency and renewable integration but increases electricity demand and cyber risks. Regional disparities in digital infrastructure create uneven benefits and challenges. Emerging technologies, like IoT and AI, can optimize energy use but require robust policies for equitable implementation. New Identified Themes: The dual role of digitalization in enhancing decarbonization while posing new risks like increased demand and cybersecurity threats. The necessity for tailored policies addressing regional and technological variations. What Has Been Found: Digital technologies, such as big data, IoT, and blockchain, play a critical role in transforming energy systems, but they require thoughtful integration to avoid unintended consequences like inequity and environmental strain.
Contributions to knowledge (What is the new knowledge/ How the identified knowledge gap is filled?)	The study provides a comprehensive framework for understanding the intersection of digitalization and decarbonization, offering actionable insights for policymakers to optimize digital technologies while mitigating risks.
Limitations (possibility for future research, still existing knowledge gap)	Limited focus on long-term economic impacts of digitalization. Uneven coverage of stakeholder perspectives across regions. Future studies could explore specific technology-driven policies in more detail.

Your own reflections (what have you learnt, what was already known to you, what was surprising, what could have been done in a better way?)	Learned: Digitalization can significantly advance decarbonization but needs strategic management to avoid trade-offs. Surprising: The extent to which regional disparities impact the effectiveness of digital initiatives. Improvements: Broader stakeholder inclusion and sector-specific case studies would enhance practical applicability.

Feature of the article	**Your comment**
Title of the article	How Key-Enabling Technologies' Regimes Influence Sociotechnical Transitions: The Impact of Artificial Intelligence on Decarbonization in the Steel Industry
Author(s)	Nikhil John, Joeri Hendrik Wesseling, Ernst Worrell, Marko Hekkert
Name of the journal/conference	Journal of Cleaner Production
Research motivation stated by the author(s) (Why is the research important? What is the knowledge gap?)	The steel industry faces major challenges in decarbonization due to its reliance on fossil fuels and existing technologies. This study explores how artificial intelligence (AI), a key enabling technology (KET), can support low-carbon innovation and transition the industry towards sustainability.
Objective(s) & research question(s) (What is going to be done?)	The research examines the technical and sociotechnical impact of AI in the steel industry, focusing on whether AI reinforces existing practices or drives transformative change towards decarbonization.
Concepts and theories used	Key-Enabling Technologies (KETs): AI as a driver of innovation and sustainability. Multi-Level Perspective (MLP): Understanding the interaction between existing regimes and niche innovations. Barriers to Innovation: Exploring challenges like path dependency, risk aversion, and system integration.
Methodology, data collection and data analysis (How was the research conducted?)	Inventory Method: Cataloged AI applications in the steel industry from scientific literature. Interviews: Conducted with industry experts to analyze AI's impact on innovation barriers.

	Data Analysis: Thematic coding of inventory and interview data to identify patterns and insights.
Key findings or results/ New identified themes (what has been found by the author(s)?)	Key Findings: AI is primarily used to improve existing technologies rather than advancing novel low-carbon solutions. It offers incremental improvements through optimization and automation but reinforces industry path dependencies. AI tools can address integration challenges and improve flexibility, aiding in scenario planning and risk management. New Identified Themes: AI's dual role: It supports both regime stability and niche innovations. Data availability and institutional inertia significantly influence AI adoption in low-carbon applications. What Has Been Found: AI applications are more focused on efficiency and cost-effectiveness rather than transformative decarbonization. While some AI tools aid low-carbon technologies, their potential remains underutilized.
Contributions to knowledge (What is the new knowledge/ How the identified knowledge gap is filled?)	The study highlights the nuanced role of AI as both a stabilizer of existing systems and a potential enabler for low-carbon transitions. It provides a framework for assessing KET impacts on sustainability transitions.
Limitations (possibility for future research, still existing knowledge gap)	Limited generalizability beyond the steel industry.

	Focused more on technical impacts, with less exploration of broader societal and policy dimensions. Future research could explore long-term effects of AI in other sectors and regions.
Your own reflections (what have you learnt, what was already known to you, what was surprising, what could have been done in a better way?)	Learned: AI's potential to reduce emissions lies in its ability to optimize and innovate across industrial processes. Surprising: The industry's reliance on AI to enhance existing technologies rather than disrupt them. Improvements: A broader study across multiple sectors could reveal generalizable insights into AI's role in decarbonization.

Feature of the article	**Your comment**
Title of the article	Energy Democracy: A Digital Future?
Author(s)	Emily Judson, Oscar Fitch-Roy, Iain Soutar
Name of the journal/conference	Energy Research & Social Science
Research motivation stated by the author(s) (Why is the research important? What is the knowledge gap?)	The authors explore the intersection of digitalization and energy democracy. They highlight how digital advancements influence citizen participation, governance, and energy ownership while addressing the knowledge gap in how digitalization reshapes democratic principles in the energy sector.
Objective(s) & research question(s) (What is going to be done?)	The study aims to examine how digital technologies affect energy democracy's core dimensions—popular sovereignty, participatory governance, and civic ownership—and whether digitalization enhances or undermines democratic energy practices.
Concepts and theories used	Energy Democracy Framework: Focus on citizen engagement, governance inclusivity, and ownership. Sociotechnical Transitions: Interplay between technology and social change. Digitalization Impacts: Examining challenges like data control, transparency, and automation.
Methodology, data collection and data analysis (How was the research conducted?)	The research uses a conceptual review, analyzing literature on energy democracy and digitalization. It evaluates theoretical insights and emerging case studies to highlight evolving trends and implications.
Key findings or results/ New identified themes	Key Findings: Digitalization can empower citizens by offering more ways to participate but also complicates

(what has been found by the author(s)?)	accountability due to automation and algorithmic decision-making. Inequalities in digital access and skills risk excluding some groups from meaningful participation. The concept of ownership is evolving to include digital assets, but dependencies on proprietary software raise challenges. New Identified Themes: Digitalization's dual role: It creates opportunities for democratic engagement but also introduces risks like exclusion and loss of transparency. Shifting definitions of ownership in a digitalized energy sector. What Has Been Found: Digitalization is reshaping energy democracy, expanding citizen roles while introducing complexities in governance, accountability, and inclusivity. The interplay between data, software, and material assets demands new democratic frameworks.
Contributions to knowledge (What is the new knowledge/ How the identified knowledge gap is filled?)	The paper provides a fresh perspective on how digital technologies interact with energy democracy, advocating for new tools and policies to address challenges while maximizing the benefits of digitalization.
Limitations (possibility for future research, still existing knowledge gap)	Focuses on theoretical and early-stage case studies, limiting empirical insights. The study mainly addresses industrialized contexts, which might not apply to other regions. Future research could explore practical policy implementations and global comparisons.

Your own reflections (what have you learnt, what was already known to you, what was surprising, what could have been done in a better way?)	Learned: Digitalization holds immense potential to redefine democracy in energy systems. Surprising: The degree to which automation complicates accountability and governance. Improvements: Empirical studies across diverse regions and contexts would enhance the applicability of findings.

Feature of the article	**Your comment**
Title of the article	The Role of Powerful Incumbent Firms: Shaping Regional Industrial Path Development through Change and Maintenance Agency
Author(s)	Simon Baumgartinger-Seiringer
Name of the journal/conference	Regional Studies, Regional Science
Research motivation stated by the author(s) (Why is the research important? What is the knowledge gap?)	The research investigates how powerful incumbent firms influence regional industrial path development, balancing between maintaining stability and driving change. It fills a gap in understanding how these firms use their power to either sustain existing systems or innovate.
Objective(s) & research question(s) (What is going to be done?)	The study explores how incumbent firms exert influence, focusing on their dual role in fostering industrial digitalization while resisting decarbonization, using Austria's automotive industry as a case study.
Concepts and theories used	Evolutionary Economic Geography (EEG): Examining industrial change. Regional Innovation Systems (RIS): Interplay of structures and agency. Change and Maintenance Agency: Strategies for innovation or stability.
Methodology, data collection and data analysis (How was the research conducted?)	Conducted 25 in-depth interviews with industry, government, and academic stakeholders. Analyzed 140 documents, including policy reports and company statements. Used qualitative content analysis to explore themes of power and influence.
Key findings or results/	Key Findings:

New identified themes (what has been found by the author(s)?)	Incumbents used their power to maintain combustion engine technologies, resisting decarbonization. They actively supported digitalization through collaborations, funding, and influencing education systems. Maintenance agency manifested through lobbying, resource allocation, and discursive strategies. New Identified Themes: The dual role of incumbents in balancing change and stability. Influence of regional and extra-regional dynamics on industrial development. What Has Been Found: Incumbents act as both barriers and enablers of regional industrial transitions, leveraging various forms of power to shape economic and innovation trajectories.
Contributions to knowledge (What is the new knowledge/ How the identified knowledge gap is filled?)	The study advances understanding of the nuanced roles incumbents play in industrial transitions. It highlights the importance of considering power dynamics in shaping regional innovation systems.
Limitations (possibility for future research, still existing knowledge gap)	Focuses on Austria, limiting broader applicability. Data covers a specific timeframe (2012-2019), which may not capture recent trends. Calls for further studies in less-developed regional systems.
Your own reflections (what have you learnt, what was already known to you, what was surprising, what could	Learned: Incumbents wield significant influence in shaping industrial paths, often resisting sustainability for economic stability.

have been done in a better way?)	Surprising: The extent to which digitalization efforts are supported while decarbonization is resisted. Improvements: Including broader comparative studies could provide more generalizable insights.

Annex 4. The conceptual framework driven from the study

Dimensions	Themes	Sub-Themes	Short Descriptions with References
Technological	Solar PV Performance	Dust Accumulation	Prolonged dust accumulation significantly impacts solar PV performance and efficiency, highlighting the need for maintenance (Ahmad et al., 2024).
		Maintenance	Cleaning and upkeep are crucial for sustaining PV efficiency and energy output (Ahmad et al., 2024).
		Off-Grid Optimization	Optimizing load management can enhance performance in off-grid solar PV systems (Ahmad et al., 2024).
	Construction Digitalization	Prefabrication	Prefabricated components paired with digital tools improve energy performance and reduce construction errors (Piaia et al., 2024).
		Real-Time Insights	AR and BIM provide workers with actionable, real-time information, bridging gaps between design and execution (Piaia et al., 2024).
	Data Centers & Energy Systems	Waste Heat Recovery	Recovering waste heat from data centers can meet significant portions of heating demands, aiding decarbonization (Li et al., 2024).
		Sector Coupling	Synergies between sector coupling and energy optimization promote sustainable systems (Li et al., 2024).
	Infrastructure Maintenance	Precision Maintenance	BIM enables precision in maintenance, reducing emissions and enhancing performance (de Bortoli et al., 2024).
		Monitoring	Real-time pavement monitoring addresses gaps in traditional road maintenance methods (de Bortoli et al., 2024).

	Artificial Intelligence	Dual Role of AI	AI supports both stability in existing regimes and innovations for low-carbon applications (John et al., 2024) .
		Institutional Barriers	Data availability and organizational inertia limit AI's adoption in transformative ways (John et al., 2024).
Organizational	Organizational Digitalization	Knowledge Management	"Single-use knowledge" drives carbon emissions, underscoring the importance of cultural shifts toward sharing and sustainability (Jackson & Hodgkinson, 2024).
		Technology-Sustainability Balance	Balancing advancements with environmental responsibility is critical for long-term success (Jackson & Hodgkinson, 2024) .
	Energy Companies & Decentralization	Centralized Systems	Despite decentralization trends, centralized models remain dominant among large incumbents (Kattirtzi et al., 2024).
		Regional Strategies	Regional and parent company strategies shape energy transitions (Kattirtzi et al., 2024).
	Incumbent Strategies	Balancing Stability & Change	Incumbents balance maintaining existing systems with embracing innovation for sustainability (Baumgartinger-Seiringer, 2024).
		Regional Dynamics	Both regional and extra-regional factors influence industrial development and digitalization strategies (Baumgartinger-Seiringer, 2024).
Societal	Public Administration	Digital Participation Platforms	Platforms play a critical role in decarbonization but are hindered by insufficient infrastructure (Jakimowicz & Rzeczkowski, 2024).
		Renewable Energy in Rural Areas	Small-scale renewable energy projects can benefit rural regions, fostering sustainable development (Jakimowicz & Rzeczkowski, 2024).

	Societal Impacts of Digitalization	Energy Sufficiency & Skepticism	Societal values shape digitalization pathways, often reflecting skepticism toward technological fixes (Gantioler et al., 2024).
		Digital Divides	Vulnerabilities like energy poverty are exacerbated by unequal access to digital tools (Gantioler et al., 2024).
	Energy Democracy	Citizen Engagement	Digital tools enable democratic participation but risk exclusion and loss of transparency (Judson et al., 2024).
		Ownership Redefinition	Ownership in the energy sector is shifting with the integration of digital assets and platforms (Judson et al., 2024).